COM102 – Programming Language C

Exercise

1. Develop a program to check whether a number is divisible by both 3 and 5. If the number is divisible by both, print "Divisible by 3 and 5". Otherwise, print "Not divisible by both.

2. Formulate a program that repeatedly asks the user to input positive numbers and adds them to a sum. The program should stop asking for input when the user enters a negative number. Once the loop ends, print the total sum of the numbers entered.

3. Simulate a basic ATM withdrawal system. The user starts with a balance of $1000. Write a program that repeatedly asks the user how much they want to withdraw and subtracts it from the balance. If the withdrawal amount is greater than the current balance, print "Insufficient funds" and ask again. The loop should stop when the balance reaches zero.

4. Write a program that calculates the squares and cubes of the numbers from 0 to 10 and uses tabs to print the following table of values:

```
number  square  cube
0       0       0
1       1       1
2       4       8
3       9       27
4       16      64
5       25      125
6       36      216
7       49      343
8       64      512
9       81      729
10      100     1000
```

5. Develop a program that reads a student's grade (0-100) and displays the corresponding letter grade using `if-else-if` statements:
   - A: 90-100
   - B: 80-89
   - C: 70-79
   - D: 60-69
   - F: 0-59

Requirements: Include validation for out-of-range grades.

6. Create a simple calculator that takes two numbers and an operator (+, -, *, /) as input and displays the result using a switch statement.
   Requirements: Handle division by zero and invalid operators.

7. Write a program that calculates the factorial of a given positive integer using a for loop.
   Requirements: Include error handling for negative inputs.

8. Develop a program that prompts the user for a number and prints its multiplication table using a while loop.
   Requirements: Allow the user to choose how many multiples to display.

9. Create a program that prints a right triangle of asterisks (*) based on the number of rows input by the user using nested loops.
   Requirements: Prompt the user for the number of rows and ensure it is a positive integer.

10. Write a program that checks if a number entered by the user is a prime number. Use a loop to check for factors and break when a factor is found.
    Requirements: Implement input validation to only accept positive integers greater than 1.

11. Develop a program that prints all odd numbers from 1 to 100 using a for loop and the continue statement to skip even numbers.
    Requirements: Clearly indicate the output format.

12. Write a program that reads a positive integer from the user and calculates the sum of its digits using a while loop.
    Requirements: Use the conditional operator to check if the digit is odd or even while summing

13. Create a simple number guessing game where the program generates a random number between 1 and 100. The user has to guess the number, and the program should provide hints (too high, too low) using if-else statements. Use a break statement to exit the loop when the user guesses correctly.
    Requirements: Limit the number of attempts to 10 and display the total attempts made by the user.

14. Write a program that takes an integer input from the user and determines whether the number is odd or even using an if-else statement.
    Requirements: Use input validation to ensure the user enters a valid integer.